1. Line Graph (2 Questions)

Question 1

Create a **line graph showing the trend of average subject scores (math, reading, writing)** across different **parental levels of education**.

Requirements

- Convert the dataset from wide to long format.
- Calculate the **mean score per subject per parental education level**.
- Plot a **multi-line chart** where each subject is a separate line.

Analysis Task

- Which subject shows the strongest improvement with higher parental education?

Question 2

Construct a **line graph comparing the average scores of students who completed the test preparation course vs those who did not**, across the three subjects.

Requirements

- Calculate average score per subject grouped by `test preparation course`.
- Convert the data to long format.
- Use `color` to distinguish preparation groups.

Analysis Task

- Does the preparation course improve performance consistently across subjects?

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2. Bar Graph (2 Questions)

Question 3

Create a **grouped bar chart showing the average math score by gender and test preparation course**.

Requirements

- Use `group_by(gender, test preparation course)`
- Calculate average math score.
- Display bars side-by-side.

Analysis Task

- Which group shows the highest average math score?

Question 4

Create a **stacked bar chart showing the distribution of students across parental education levels by lunch type**.

Requirements

- Count students per category.
- Use `fill = lunch`.

Analysis Task

- Which parental education group has the highest proportion of students receiving free/reduced lunch?

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3. Scatter Plot (2 Questions)

Question 5

Create a **scatter plot showing the relationship between math score and reading score**.

Requirements

- Color points by `gender`.
- Size points by `writing score`.
- Add a **regression line using `geom_smooth(method = "lm")`.**

Analysis Task

- Does gender influence the correlation between math and reading scores?

Question 6

Create a **scatter plot examining the relationship between writing score and reading score**, faceted by `test preparation course`.

Requirements

- Add regression lines for each facet.
- Use transparency (`alpha`) to reduce overplotting.

Analysis Task

- Does preparation course affect the strength of the relationship?

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4. Histogram (2 Questions)

Question 7

Create a **histogram showing the distribution of math scores**.

Requirements

- Use 15–20 bins.
- Facet the histogram by `gender`.

Analysis Task

- Are there differences in score distribution between male and female students?

Question 8

Create a **histogram of writing scores grouped by lunch type**.

Requirements

- Use `fill = lunch`.
- Use `position = "identity"` with transparency.

Analysis Task

- Do students receiving free/reduced lunch show a different score distribution?

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5. Advanced Histogram (Faceted Analysis)

Question 9

Create a **faceted histogram of reading scores across parental education levels**.

Requirements

- Use `facet_wrap(~ parental level of education)`
- Compare distribution shapes.

Analysis Task

- Which education category has the most **right-skewed or left-skewed distribution**?

Question 10

Create a **histogram showing the distribution of average student score**.

Requirements

- First create a new variable:

```
average_score = (math score + reading score + writing score)/3
```

- Plot the histogram.
- Add a vertical line showing the mean score.

Analysis Task

- Is the distribution approximately normal?